DWIJ VYAS

Electronics & Communication Engineering — Embedded Systems, Robotics & Applied AI

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SUMMARY

I'm an Electronics & Communication Engineering undergraduate at BVM, working across embedded systems, digital electronics and applied AI. My work spans hardware design, firmware and the software that ties them together. I'm looking for an internship where I can do this kind of work on a real team.

EDUCATION

Birla Vishvakarma Mahavidyalaya *'24 — now*

B.Tech, Electronics & Communication · Vallabh Vidyanagar · CPI 9.05 / 10

Bright Day School *'23 — '24*

Higher Secondary, Science · Vadodara · 87.8%

TECHNICAL SKILLS

Embedded: C, C++, ESP32 / ESP8266, ATmega328P, Arduino, Arduino IDE, Sensor fusion, Wireless communication

Robotics: Pixhawk 2.4.8, Raspberry Pi, Jetson Nano, Mission Planner, QGroundControl, Autonomous navigation, Robotics

Digital logic: Verilog, BASYS3 FPGA, Xilinx Vivado, Digital electronics, CPU architecture

Software & AI: Python, MATLAB, C++, Linux / Ubuntu, Autonomous navigation, Sensor fusion

Fabrication: Fusion 360, EasyEDA, PCB design, Bambu Studio, 3D modelling

PROJECTS

TerraAvis — Hybrid drone–tank rescue robot

Robotics · Embedded AI · Pixhawk

An AI-enabled hybrid rescue robot with a dual-mode mobility system combining aerial navigation and tracked ground movement. Integrated thermal imaging, FPV surveillance, real-time object detection and autonomous control for search-and-rescue and disaster-response scenarios.

Technologies: Pixhawk, Raspberry Pi, Python, OpenCV, Thermal imaging, PCB design, Fusion 360

Documentation: https://drive.google.com/drive/folders/1DYj1aq4hzN4QaE1onNEsXEsNY5stDXUG?usp=sharing

4-Bit Microprocessor — Built from discrete logic on breadboards

Digital electronics · CPU design

A fully functional 4-bit microprocessor built entirely from discrete 74-series logic ICs on breadboards. Complete datapath including ALU, registers, instruction register, program counter and shared bus, with a hardwired control unit and a custom instruction set supporting arithmetic, logical, load, store and conditional jump operations.

Technologies: 74-series TTL, 555 timer, Logisim, Proteus, Breadboard prototyping

Documentation: https://drive.google.com/drive/folders/1dfLMLjH02TEv6RGfnhIRi95xKUdKb2f0?usp=sharing

Friendly AI Chatbot — Embedded voice assistant

Embedded hardware · AI · PCB design

A standalone embedded AI device built around an ESP32 with real-time voice input, amplified audio output and an OLED display for feedback. Includes a custom-designed PCB and 3D-printed enclosure, with Python driving the conversational logic.

Technologies: ESP32, OLED SSD1306, PAM8403, EasyEDA, Python, Arduino IDE

Documentation: https://drive.google.com/drive/folders/1A3rikUEOEEWcZX6W_JjXLqyuPbLvr2Cd?usp=sharing

Dual-Axis CNC Plotter — Motion control from G-code to paper

Embedded systems · Mechanical design

A 2-axis CNC plotter using Arduino, stepper motors and a custom mechanical frame to translate digital designs into precise physical drawings. Implemented G-code interpretation and motion-control firmware for accurate pen positioning, combining embedded systems with mechanical design.

Technologies: Arduino, GRBL firmware, Inkscape, Stepper motors, C/C++, Fusion 360

RECOGNITION & EXPERIENCE

Robofest 5.0 Gujarat — Finalist Recognition · 2025
TerraAvis, a hybrid drone–tank rescue robot, reached the state finals.